DE–4133 **11**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

CLASSICAL AND STATISTICAL MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
Each question carries 20 marks.

(5 × 20 = 100)

1. (a) What is meant by generalized coordinates? Express its connection with the degrees of freedom. (5)

   (b) Derive Lagrangian equation of motion from the principle of virtual work. (10)

   (c) Derive Lagrangian equation of motion from the variational principle. (5)

2. (a) When a transformation is said to be canonical? Give examples. (5)

   (b) Write down the properties of Poisson brackets. (5)

   (c) Solve the Kepler problem using action angle variables. (10)

3. (a) What is the significance of cyclic coordinates? (5)

   (b) Obtain Hamilton's equations in Poisson bracket form. (10)

   (c) Distinguish between conservative and non-conservative systems. (5)

4. (a) Define the normal coordinates and normal modes with examples. (5)

(b) Discuss about Euler angles. (5)

(c) Obtain the normal modes for linear triatomic molecule. (10)

5. (a) Derive Euler equations. (10)

(b) Study the motion of symmetric top under the influence of gravity. (10)

6. (a) State and Prove Liouville's theorem. (10)

(b) What is meant by partition function? Explain its importance in statistical mechanics. (5)

(c) Study about the classical ideal gas in canonical ensemble. (5)

7. (a) Apply Maxwell Boltzmann statistics and obtain expression for velocity distribution. (10)

(b) Deduce expressions for r.m.s and most probable velocities. (5)

(c) Write down the disadvantages of Maxwell Boltzmann statistics. (5)

8. Write notes on :

(a) Hamilton-Jacobi theory. (7)

(b) Angular momentum of rigid body. (7)

(c) Equipartition of energy. (6)

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

MATHEMATICAL PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) State and prove Cayley-Hamilton theorem.

(b) State Schwarz inequality and describe Gram-Schmidt's orthogonalization process. (10)

(c) Show that the eigen values of Hermitian matrix are all real. (5)

2. (a) Find all the eigen values and eigen vectors of the matrix $M = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$ (8)

(b) Show that $\overline{|n+1} = N\overline{|n}$. (5)

(c) Show that $\int_{-1}^{1} p_M(\mu)\, P_n(\mu)\, d\mu = \frac{2}{2n+1}\delta_{m,n}$ where $m, n > 0$. (7)

3. (a) Solve the Hermite equation

$$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2ny = 0$$

when n is a parameter. (10)

(b) Prove that $H_n(-x) = (-1)^n H_n(x)$. (5)

(c) Show that if m is an integer $\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx = 0$.(5)

4. (a) Prove that $L\left[e^{at}f(t)\right] = F(s-a)$. (5)

(b) Using Parseval's identity, prove $\int_0^{\infty}\left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}$.

(5)

(c) State and prove Cauchy-Riemann integral theorem.

(10)

5. (a) State and prove Cauchy-Residue theorem. (8)

(b) Apply Calculus of residues to show that $\int_0^{2\pi}\frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}$, $a > b > 0$. (8)

(c) Find the Laplace transform of t^n. (4)

6. (a) Discuss the vibration of a rectangular membrane.

(4)

(b) Solve the two dimensional heat flow equation. (8)

(c) Discuss the importance of Laplace equation and obtain its solution. (8)

7. (a) Explain the moment of inertia tensor of rigid body. (8)

(b) Is force a tensor? If so, what is its nature? Explain with reason. (6)

(c) Define contraviant, covariant tensors. Give examples. (6)

8. (a) Define isomorphic and homomorphic groups. Differentiate between them. (4)

(b) Explain the concept of group representation. Prove the orthogonality theorem in group theory. (8)

(c) Construct the symmetry group of a square (D_4). Write down its multiplication table, classes, subgroups. (8)

DE–4135 **13**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

INTEGRATED AND DIGITAL ELECTRONICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Explain class A power amplifier. How power is distributed in it. (6)

   (b) Explain the 'black box representation' of a transistor. Write down the hybrid equations of a transistor. (8)

   (c) Derive the general expression for the collector efficiency of a power amplifier. (6)

2. (a) Explain the working of source follower amplifier in a FET. Calculate its voltage gain. (10)

   (b) Define and explain the parameters of FET. What is the relation among them? (8)

   (c) Define Miller theorem. (2)

3. (a) List the characteristics of an ideal op-amp. (5)

   (b) Draw the schematic diagram and derive the expression for the output voltage for an ideal non inverting op-Amp. (7)

   (c) Draw the op-Amp circuit of a voltage to current converter. Explain its operation. Mention any one use of this circuit. (8)

4. (a) Explain the working of op-amp as (10)

(i) Differentiator

(ii) Comparator

(b) Explain how the simultaneous equation is solved by op-amp with suitable example. (10)

5. (a) What is Boolean Algebra? Explain the Boolean algebraic operations and state the rules for such operations. (10)

(b) Minimize the following functions and realize using minimum number of gates. (10)

$$f = \sum m(0, 3, 5, 6, 9, 10, 12, 15)$$

6. (a) State and prove De Morgan's theorem (6)

(b) What is an RS-Flipflop? Give its truth table. Explain how a Flip-Flop can be obtained by using two single-input NAND gates. (7)

(c) Explain the operation of Mod-n-counters (7)

7. (a) Explain the internal architecture of 8085 μp and discuss the function of different blocks. (10)

(b) Write a note on

(i) Fetch operation

(ii) Execute operation (5)

(c) Discuss the addressing modes in 8086 with examples. (5)

8. (a) Write a detailed account on fabrication of monolithic IC's. (8)

(b) Write a note on Hall effect (4)

(c) Explain the Fabrication of Solar Cells. (8)

DE–4136 **14**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

ELECTRO MAGNETIC THEORY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Write a note on maxwell's equation in free space. (8)

   (b) Explain Maxwell's equations for harmonically varying fields. (7)

   (c) Explain how maxwells equation can be used to determine the boundary surface of two different media. (5)

2. (a) Explain the total internal reflection at an interface between two dielectrics. (6)

   (b) Give the kinetic properties in reflection and refraction of plane electromagenetic wave. (8)

   (c) Explain Poynting theorm and Poynting vector. (6)

3. (a) Explain the working of a klystron with a neat diagram. (8)

   (b) What do you mean by resonant cavities? (7)

   (c) Write a note on TEM modes. (5)

4. (a) With a neat diagram explain the working of a magnetron. (8)

(b) What is a gunn diode? (6)

(c) Write a note on rectangular wave guide. (6)

5. (a) What is dispersion (Normal and Anomalous)? Explain the dispersion in Gases, solides and liquids. (10)

(b) What is Lorentz formula? (6)

(c) Write a note on polarization on scattered light. (4)

6. (a) Derive the clasusius massatti relation. (8)

(b) Write a note on scattering and scattering parameters. (5)

(c) Give the theory of scattering of electromagnetic wave. (7)

7. (a) State any two application of plasma. (5)

(b) Give the condition for plasma existence and occurrence. (8)

(c) Write a note on magneto hydrodynamics. (7)

8. (a) Give the boundary condition at the surface of the discontinuity. (8)

(b) Derive differential form of maxwell's equation. (8)

(c) Distinguish isotropic and an isotrophic medium. (4)

DE–4137 **15**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

NUMERICAL METHODS AND PROGRAMMING

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Discuss the four basic data types. How could we extend the range of values they represent? (7)

   (b) Write a C program to solve the quadratic equation $ax^2 + bx + c = 0$, where a, b, c are constants. (8)

   (c) Explain the purpose of scanf ( ) function. (5)

2. (a) Give the rules for declaring variables in C. (5)

   (b) Discus arithmetic operators and the rules of precedence. (7)

   (c) Describe various types of if statements available in C. (8)

3. (a) What is recursion? Explain with suitable example the working of a recursive function. (8)

   (b) Describe the syntax for the declaration of structure. (5)

   (c) Explain the various steps involved in opening a data file to read data from it. (7)

4. (a) Describe Newton-Raphson method to find the roots of given function and emergence of it. (10)

(b) Find a root of the equation $x^3 - 4x - 9 = 0$ correct to three decimal places using the bisection method. (5)

(c) Using Newton-Raphson method, establish the formula $x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_0}\right)$ to calculate the square root of N. (5)

5. (a) Use the method of least squares to fit a straight line to the following data : (8)

x :	0	5	10	15	20
y :	7	11	16	20	26

Estimate the value of y when $x = 25$.

(b) Obtain trapezoidal rule. (4)

(c) Use Runge-Kutta method to find the numerical solution at $x = 0.8$ for

$\frac{dy}{dx} = \sqrt{x + y}$, $y(0.4) = 0.41$. Assume the step length $h = 0.2$. (8)

6. (a) Explain Gauss-Seidal method of iteration. (7)

(b) Solve, by Gauss-Jacobi method the equations : (8)

$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72$$
$$x + y + 54z = 110$$

(c) Describe the method of least squares. (5)

7. (a) Derive Simpson's 1/3 rule. (5)

(b) Describe a second order Runge-Kutta formula to solve first differential equations. (7)

(c) Discuss about the supremacy of bisection method. (8)

8. (a) List the difference between the while and do-while looping statements. (5)

(b) Explain how multiple values are returned from the functions. (8)

(c) Explain the two different ways to access a array element. (7)

DE–4138 **21**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

SPECTROSCOPY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Explain molecular orbital theory. (6)

   (b) Explain Heitler London theory for hydrogen molecule. (6)

   (c) Describe the terms SP^2 and SP^3 hybrid orbitals. (8)

2. (a) Explain with suitable examples and diagram-symmetric top molecules. (6)

   (b) Describe the action of a diatomic molecule as a rigid rotator. (8)

   (c) What is Raman affect? Give the conditions to be Raman active. (6)

3. (a) What is stark effect? Give its importance in microwave spectroscopy. (7)

   (b) State and explain mutual exclusion principle. (6)

   (c) Why both IR and Raman spectroscopy are essential tools for the study of molecular structure. (7)

4. (a) Describe in detail about the symmetry and fundamental vibrations of diatomic molecules. (7)

   (b) Explain stoke's Raman scattering. (6)

   (c) Describe the photo acoustic Raman scattering. (7)

5. (a) Describe the vibrational spectra of polyatomic molecules. (8)

(b) State and explain inverse Raman effect. (6)

(c) Write a note on predisassociation. (6)

6. (a) Explain in detail the pure rotation Raman spectra. (7)

(b) Explain the principle of ESR and appearance of hyperfine structure. (7)

(c) What are the causes of chemical shift? Explain. (6)

7. (a) Explain the principle of NQR. (5)

(b) Derive Block equation for nuclear resonance. (10)

(c) Discuss the applications of NMR. (5)

8. (a) Explain the techniques and instrumentation of Raman Spectroscopy. (8)

(b) Explain Mossbaure spectroscopy. (6)

(c) Highlight the physics of dipole–dipole interaction. (6)

DISTANCE EDUCATION

M.Sc.(Physics) DEGREE EXAMINATION, MAY 2012.

QUANTUM MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry 20 marks.

($5 \times 20 = 100$)

1. (a) What is uncertainty principle? Give a mathematical proof of the position momentum uncertainty in quantum mechanics. (10)

   (b) What are De-Broglie waves? Discuss their properties. (5)

   (c) Write a note on probability current density. (5)

2. (a) Obtain and solve Schrodinger equation for one dimensional linear harmonic oscillator. (10)

   (b) Solve Schrödinger's equation for hydrogen atom and discuss its radial wave function. (10)

3. (a) Solve Schrodinger's wave equation for a rigid rotator with a free axis. Discuss its degeneracy and parity. (14)

   (b) Explain zero point energy in harmonic oscillator. (6)

4. (a) Explain Dirac notations. (6)

   (b) Write notes on parity operator and projection operator. (14)

5. (a) Describe WKB approximation method with one application. (14)

(b) Write a note on Einstein's A and B coefficients. (6)

6. (a) Define scattering cross section. Describe the method of partial wave for scattering. (12)

(b) Explain Fermi's golden rule. (8)

7. (a) Explain Rayleigh scattering. (8)

(b) Obtain CG coefficients when two angular moments $J_1=1/2$ and $J_2 = 1/2$ are coupled. (12)

8. (a) Derive the Eigen value of operators J^2 and J_z. (14)

(b) Explain diffusion scattering. (6)

DE–4140 **23**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

SOLID STATE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

Each questions carry 20 marks.

1. (a) Explain Bragg's law for x-ray diffraction in crystals. (5)

   (b) Describe briefly the seven system of crystals. (10)

   (c) Explain with suitable examples,The ionic, covalents metallic type of bonding in solids. (5)

2. (a) What are elastic compliance constants and elastic stiffness constants? (5)

   (b) Show that in cubic crystals there are only there independent stiffness constants. (8)

   (c) Explain the following

      (i) Atomic form factor (4)

      (ii) Hydrogen bonds. (3)

3. (a) How does the Debye model differ from the Einstein model? Explain. (5)

   (b) Derive the expression for the specific heat of a solid on the Einstein temperatures it drops exponentially with decreasing temperature. (8)

   (c) Derive the expression for thermal conductivity on the basis of classical free electron theory. (7)

4. (a) What is Fermi energy? Write down its relation with the concentration of electrons in metals. (4)

(b) Explain the heat capacity of the electron gas. (10)

(c) State and explain bloch theorem. (6)

5. (a) Obtain the expression for the electrical conductivity and resistivity of metals. (10)

(b) Explain the vibrations of crystal with monatomic basic. (10)

6. (a) Explain donor states and acceptor states of semi conductor crystals. (8)

(b) How do we go from fermi surfaces for free electrons to fermi surface for nearly free electrons? (4)

(c) Obtain the equation for calculation of energy bands using weigner -seitz method. (8)

7. (a) Explain the Landan theory of the phase transition. (8)

(b) Write notes on :

(i) Antiferro electricity. (4)

(ii) Ferro electric domains (4)

(iii) Piezo electricity. (4)

8. (a) What are paramagnetic and diamagnetic materials? Give examples. (5)

(b) What is meant by local electric field at an atom? Explain. (10)

(c) Explain magnetic bubble domains. (5)

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

NUCLEAR AND PARTICLE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Discuss about Gamow's theory of α-decay. (10)

   (b) Explain the measurement of neutrino helicity. (5)

   (c) Give a short note on nuclear isomerism. (5)

2. (a) Discuss the Bohr-Wheeler theory and to explain its mechanism. (8)

   (b) What are magic number nuclei? Give its significance. (7)

   (c) Distinguish between nuclear shell model and liquid drop model. (5)

3. (a) Write a note on simple theory of deutron. (7)

   (b) Give the theory of low energy n-p scattering. (7)

   (c) Explain the compound nuclear model of nuclear reactions. (6)

4. (a) Briefly discuss about the theory of neutron cycle in a thermo nuclear reactor. (10)

   (b) Write down the sub-nuclear particles. (5)

   (c) Explain nuclear fusion with one example. (5)

5. (a) What are fundamental interactions? (5)

(b) Write a brief account on translations in space. (8)

(c) Describe CPT theorem. (7)

6. (a) Write about the single particle model. (5)

(b) Discuss in detail magnetic moments of the shell model. (7)

(c) What are the salient features of collective model of nucleus? (8)

7. (a) Explain the normalization of deutron wave functions. (5)

(b) Explain the various types of nuclear reactors. (10)

(c) Discuss the energy release in the fusion reactions. (5)

8. (a) Explain the sources of stellar energy. (5)

(b) Explain SU(2) and SU(3) symmetry groups. (10)

(c) Explain the plasma confinement. (5)

DE–4142 **25**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2012.

MATERIALS SCIENCE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Define creep: Explain its phases and mechanism. (10)

   (b) Describe addition and condensation polymerization. (10)

2. (a) What are the characteristics of laser? Describe the construction and working of Nd:YAG laser. (14)

   (b) Explain communication and engineering application of laser. (6)

3. (a) What is population inversion and how can it be achieved? (8)

   (b) Explain the working and applications of chemical laser. (12)

4. (a) Explain the working of an electro–optic modulator using ken effect. (10)

   (b) Explain the second harmonic generation in Ammonium dihydrophosphate (ADP) crystal with necessary theory. (10)

5. (a) Discuss in detail the sputtering methods and mention their advantages. (12)

(b) Explain the principle and working of penning gauge. (8)

6. (a) What are copper ion conductors explain their applications. (12)

(b) Write an essay on fuel cell. (8)

7. (a) Describe the principle, theory and working of diffusion pump. (12)

(b) Write a short note on LB films and its applications. (8)

8. (a) Explain in detail the weight gain method. (10)

(b) What is an optical resonator? Explain its purpose in lasers. (10)